§ 13 Variation

1. Classical Physics

Fermat : $\delta \int n\,ds = 0$

Hamilton : $\delta \int L\,dt = 0$

2. Dirac Principle

$$\delta \int dt \, \langle \psi(t) | i\hbar \partial_t - H | \psi(t) \rangle = 0$$

Stationary : $\delta\left(\langle \psi | E - H | \psi \rangle\right) = 0$

$$\begin{cases} \frac{\delta}{\delta \langle \psi |} \left(\langle \psi | H | \psi \rangle - E \langle \psi | \psi \rangle \right) = 0 \\ \frac{\delta}{\delta \langle \psi |} \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = 0 \end{cases}$$

e.g. Two-Level System

$$H = B_x \sigma_x + B_z \sigma_z \,, \quad |\psi_\alpha\rangle = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}$$

$$E_\alpha = \langle \psi_\alpha | H | \psi_\alpha \rangle$$

$$\frac{\partial E_\alpha}{\partial \alpha} =$$

e.g. Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$$

Guess $\psi \sim e^{-\frac{x^2}{2\sigma^2}}$

e.g. Hydrogen Atom

$$H = -\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{r}$$

Guess $e^{-\alpha r}$

Sturm Oscillation Theorem.

Virial Theorem

$$T = \frac{1}{2}\sum_i p_i \cdot v_i = \frac{1}{2}\frac{\partial H}{\partial \alpha}\xi$$

$$V(r) = g r^k$$

Scaling Transform : $\psi(r) = \alpha^{-\frac{3}{2}}\phi\left(\frac{r}{\alpha}\right)$

$$E(\alpha) = \int d\tau\ \alpha^{-3}\psi^*\left(\frac{r}{\alpha}\right)\left(-\frac{\hbar^2}{2m}\nabla^2 + g r^k\right)\psi\left(\frac{r}{\alpha}\right) \qquad \frac{r}{\alpha} = z$$

$$= \int dz\ \psi^*(z)\left[-\frac{\hbar^2}{2m}\cdot\frac{1}{\alpha^2}\nabla^2 + g(\alpha z)^k\right]\psi(z)$$

$$\frac{\partial E(\alpha)}{\partial \alpha} = \int d\tau\ \psi^*(z)\left(-\frac{\hbar^2}{2m}\frac{1}{\alpha^3}\nabla^2 + k g \alpha^{k-1} z^k\right)\psi(z)$$

$$= -\frac{2}{\alpha}\langle T\rangle + \frac{k}{\alpha}\langle V\rangle = \frac{1}{\alpha}\left(-\langle T\rangle + k\langle V\rangle\right) = 0$$

$$\Rightarrow \langle T\rangle = k\langle V\rangle$$

e.g. He Atom

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left[\phi_{1s,\uparrow}(x_1)\,\phi_{1s,\downarrow}(x_2) - \phi_{1s,\uparrow}(x_2)\,\phi_{1s,\downarrow}(x_1)\right]$$

$$H = -\frac{\hbar^2}{2m}\nabla_1^2 - \frac{\hbar^2}{2m}\nabla_2^2 + \frac{e^2}{|r_2 - r_1|}$$

$$E = \langle\psi|H|\psi\rangle$$

Improve:

① $\phi_{a,\uparrow/\downarrow} = \sum_i a_i \phi_{i,s,\uparrow/\downarrow}$, $\phi_{b,\uparrow/\downarrow} = \sum_i b_i \phi_{i,s,\uparrow/\downarrow}$

② $e^{-z(r_1 + r_2)/a_0}$, $z = \frac{Q}{e}$

Worked Examples:

1. Hartrick - Fock.

2. DFT : Density Function Theory

$$H = T + U + V$$

kinetic, interaction, single particle potential.

$$= \sum_i \frac{p_i^2}{2m} + \int \frac{n(\vec{r}_1)\, n(\vec{r}_2)}{(\vec{r}_1 - \vec{r}_2)}\, d\vec{r}_1\, d\vec{r}_2 + \int V_{ext}(\vec{r})\, n(\vec{r})\, d\vec{r}$$

H-K Theorem :

$(V(\vec{r})) \Rightarrow \psi(\vec{r}) \leftrightarrow n(\vec{r})$, 1-1 mapping for ground state $\psi(\vec{r})$

prove : (ground state)

$H = U + T + V \Rightarrow \psi(\vec{r})$

$H' = U + T + V' \Rightarrow \psi'(\vec{r})$ $\nearrow \searrow$ $n(\vec{r})$

① H :

$E = \langle \psi | H | \psi \rangle$,

$E_1' = \langle \psi' | H | \psi' \rangle = \langle \psi' | H' + V - V' | \psi' \rangle = E' + \int (V - V')\, n(\vec{r})\, d\vec{r}$

② H' :

$E' = \langle \psi' | H' | \psi' \rangle$,

$E_1 = \langle \psi | H' | \psi \rangle = \langle \psi | H + V' - V | \psi \rangle = E + \int (V' - V)\, n(\vec{r})\, d\vec{r}$

$E_2 < E_1$, $E_1' < E_1$

$\Rightarrow E_2 + E_1' < E' + E$, 矛盾.

$\Leftarrow$ Given $V(r)$, find $n(r)$, give $\psi(r)$

Target : Find $n(r)$

A single particle :

$$H_{KS} = -\frac{\hbar^2}{2m}\nabla^2 + V_{eff} \rightarrow n(r)$$

$$E = T[n(r)] + U[n(r)] + \int V_{ext}\, n(\vec{r})\, d\vec{r}$$

$$= T_{KS}[n(r)] + E_H[n(r)] + E_{XC}[n(r)] + \int V_{ext}\, n(\vec{r})\, d\vec{r}$$

$\frac{\delta E}{\delta \phi_i^*} = 0 \rightarrow$ KS Equation : ϕ: 哪來的

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V_{ext} + \int_{r'} \frac{n(r')}{|r-r'|}\, dr' + \underline{U_{xc}}\right]\phi_i = E\phi_i$$

exchange - correlation

3. BCS Theory.

在 Fermi 球上，加上两个电子对 $(k_1, -k_1)$

$$|\psi\rangle = \sum \alpha_{k_1} |k_1, -k_1\rangle$$

$$E = \langle\psi|H|\psi\rangle = \sum_k |\alpha_k|^2 \cdot 2\varepsilon_k + \left(\sum_{k,k'} V_{k,k'} \alpha_{k'}^* \alpha_k\right) \quad \text{取 Fermi 面附近} = 0$$

归一化 $\sum_k |\alpha_k|^2 = 1$

$$\min E \to \frac{\partial\left(E - \lambda \sum_k |\alpha_k|^2\right)}{\partial \alpha_k^*} = 0$$

$$\Rightarrow \quad 2\varepsilon_k \alpha_k - \sum_{k'} V_{k,k'} \alpha_{k'} = \lambda \alpha_k$$